



# SOTA INOUE

Bachelor of Engineering

## Personal Information



fromamerica@icloud.com  
si98@email.uni-freiburg.de



Freiburg im Breisgau,  
Germany



<https://orcid.org/0009-0001-1131-3913>



[www.linkedin.com/in/sota-inoue-941258260](https://www.linkedin.com/in/sota-inoue-941258260)



<https://sotainoue-2002.github.io/en/>



Nationality: Japan  
Date of Birth: January 9, 2002

## Language

- Japanese (Native)
- English (B2+ Independent- Proficient)
- Germany (B1.2, currently learning)

## Expertise

- Computational Neuroscience
- Neural Engineering
- Behavioral Decision-Making
- Observational Learning
- Neural Circuit Modeling
- Cognitive & Systems Neuroscience

## Professional Summary

Neuroscience graduate with a strong engineering background, specialized in computational modeling, reinforcement learning, and experimental neuroscience.

Experienced in Python-based data analysis, behavioral task design, and interdisciplinary research at the interface of neuroscience and engineering.

## Research Experience

### Research Assistant (HiWi), Neuropsychology Lab, University of Freiburg

*Jun 2026 – Sep 2026*

EEG, sleep, memory, and physiological signal analysis; research support for neuropsychological studies.

### Independent Research Project, Kyoto University

*Apr 2024 – Mar 2025*

Computational modeling of self-learning and observational learning using reinforcement learning, inverse reinforcement learning, and behavioral experiments.

### Research Experience, ATR Brain Function Institute

*2024*

Assisted with human fMRI experiments and gained exposure to neuroimaging data acquisition and analysis workflows.

## Education

### University of Freiburg, Germany

Master of Science in Neuroscience  
Oct 2025 – Present

### Kyoto University, Japan

Bachelor of Engineering in Electrical and Electronic Engineering  
Apr 2021 – Mar 2025

Bachelor Thesis: Computational Modeling of Decision-Making integrating Self-Learning and Observational Learning

### Kyoto University, Japan

Master of Informatics (Computational Neuroscience) – on leave  
Apr 2025 – Present

## Additional Technical Experience

- Web development (TypeScript, Node.js, React)

## Hobbies

- Independent travel, endurance training (running, calisthenics),
- football (13 years)
- podcast listening

## Teaching & Academic Engagement

### Kyoto Consortium for Japanese Studies (KCJS)

Support Instructor (Sep 2022 – Sep 2024)

- Supported international students in Japanese language learning and academic adaptation.

### Kyoto University International Summer Program (KSP)

Participant (2022), Organizer (2023)

- Contributed to the organization of an international academic program.

### Vietnam National University, Hanoi (ULIS & USSH)

Spring Vietnamese–Japanese Intercultural Program (Feb – Mar 2023)

- Led a student team in an intercultural academic presentation.

## Research Methods

### Neuroscience & Experimental Methods

- Neural signal processing (membrane to systems level)
- Neuronal signaling and synaptic plasticity
- Sensory and motor systems
- Behavioral and cognitive task design
- Psychophysics experiments (PsychoPy)
- EEG and physiological signal analysis

### Computational & Analytical Methods

- Computational modeling of learning and decision-making
- Reinforcement Learning (RL) and Inverse Reinforcement Learning (IRL)
- Neural circuit analysis and conceptual modeling
- Time–frequency analysis of neural signals
- Statistical analysis and data visualization

### Technical Skills

- Programming: Python, C
- Scientific computing and data analysis
- Machine learning and reinforcement learning
- Tools: Linux (Ubuntu), Git, Docker, Arduino

## Internship

### System Engineer Intern

#### Nocnum (Deep-Tech Startup), Japan — Apr 2024 – Present

- Developed and maintained frontend and backend systems for IoT-based water purification monitoring using TypeScript, Node.js, React, and Docker.
- Implemented sensor-based data acquisition systems, integrating electronic hardware and software components.
- Conducted data analysis on water quality sensor data and led the implementation of clustering-based analysis pipelines using Python and machine learning methods.
- Developed a domain-specific RAG (LLM) system for wastewater treatment management by integrating technical documentation, academic literature, operational know-how, and sensor data to support expert decision-making.

## Honors and Awards

### DAAD Master's Scholarship (2025–2027)

#### German Academic Exchange Service (DAAD)

- Highly competitive scholarship awarded by the German Federal Foreign Office for outstanding academic performance and research potential.

### TA Award, Electronics Spring Camp (2022 Spring)

#### Kyoto University, Department of Electrical and Electronic Engineering

- Awarded for outstanding performance in team-based electronic system design using LEGO Mindstorms.

### Grand Prize (Iwakabe Special Award), Electronics Summer Camp (2022 Summer)

#### Kyoto University, Department of Electrical and Electronic Engineering

- Led a team to develop an **Arduino**-based line-tracing robotic system and received the highest award for electronic system design.